RNC.

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name: **Renew Coil Engineering & Supply Co., Ltd.**

WPS No.: **KWC-06A0017-WPS-RNC-PN1** Date: **23-Feb-13** Revision No.: 1

Supporting PQR No.(s): **KWC-06A0017-PQR-RNC-PN1** Date: **23-Feb-13** Type(s): Manual

Welding Process(es): Gas Tungsten Arc Welding + Shield Metal Arc welding **(GTAW +SMAW)** (Automatic, **Manual**, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design: **GROOVE & FILLET**

Backing (Yes) SMAW (No)

Backing Material (Type): Weld Metal

■ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. 1 Group No. 1 or 2 to P-No. 1 Group No. 1 or 2

Material Specification Type and Grade: A106 Gr.B , A105 Gr.B, A 234 WPB (or as per P-1) TO A106 Gr.B , A105 Gr.B, A 234 WPB (or as per P-1)

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal : Groove 5 mm -20 mm. Fillet All

Pipe Dia. Range : Groove NPS $2^{7}/8$"-Unlimited Fillet All

Other: Maximum Pass Thickness 1/2 in. (13 mm) for SMAW

FILLER METALS (QW-404)

	GTAW	SMAW
Spec. No. (SFA)	5.18	5.1
AWS No. (Class)	ER 70S-G	E 7016
F-No.	6	4
A-No.	1	1
Size of Filler Metals	Ø 2.4 mm	Ø 3.2 & 4.0 mm.
Deposited Weld Metals	3.0 mm.	7.97 mm.
Thickness Range :		
Groove	6 mm Max.	15.94 mm. Max.
Fillet	ALL	ALL
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	Kobelco TGS-50 or Equivalents	Kobelco LB 52 or Equivalents

POSITION (QW-405)

Position(s) of Groove: ALL

Welding Progression: Up X Down

Position(s) of Fillet: ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range: N/A

Time Range: N/A

Notes: N/A

PREHEAT (QW-406)

Preheat Temp. Min. 30°C (Ambient)

Interpass Temp. Max. 250^{0}C

Preheat Maintenance: N/A

Notes

GAS (QW-408)

	Percent Composition		
	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	8-12 Ltr/Min
Trailing	-	-	-
Backing	-	-	-

Sheet 1 of 2

RNC.

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. KWC-06A0017-WPS-RNC-PN1 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current [X] AC [X] DC Polarity EN(GTAW)+ EP(SMAW)

Amps (Range) 94-150(GTAW) + 100-112(SMAW) Volts (Range) 10-12(GTAW)+ 23-24(SMAW)

Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)

(Pure Tungsten, 2% Thoriated, etc.)

Mode of Metal Transfer for GMAW N/A

(Spray arc, short circuiting arc, etc.)

Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS

Orifice or Gas Cup Size 8.0 mm - 19.0 mm

Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING

Method of Back Gouging N/A

Oscillation N/A

Contact Tube to Work Distance N/A

Multiple or Single Pass (per side) MULTIPASS

Multiple or Single Electrodes SINGLE ELECTRODE / MULTIPLE ELECTRODE

Travel Speed (Range) 4.73-12 (GTAW)+5-15.24(SMAW)Cm/Minute

Peening N/A

Other N/A

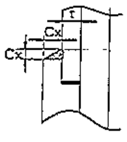
Weld Layer(s)	Process	Filler Metal Class	Filler Metal Dia. (mm)	Current Type Polar.	Current Amp. Range	Volt Range	Travel Speed Range (Cm/min)	Other
1st	GTAW	ER 70S-G	2.4	DC EN	94-110	10-11	4.73-12	
2nd	GTAW	ER 70S-G	2.4	DC EN	120-150	10-13	10-12	
3 & Over	SMAW	E 7016	3.2 & 4.0	DC EP/AC	100-112	23-34	5-15.24	

We,the undersigned, certify that the statements in this record are correct and that the test welds were prepared ,welded, and tested in accordance with the requirement of **ASME-IX, 2010 latest Revision.**

	Name	Signed	Date
Prepared by :	Mr. Chatchawan S		23/02/13
Reviewed by :	ภูชิน สีหอมไชย Phuchin Seehomchai		11/03/13
Approved by:	ภูชิน สีหอมไชย Phuchin Seehomchai		11/03/13.

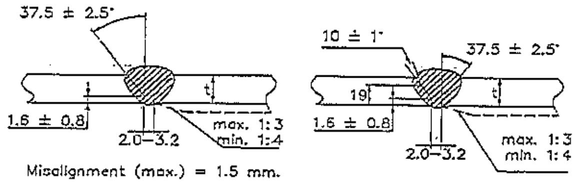
Sheet 2 of 2

ATTACHED FIGURE JOINT SKETCH



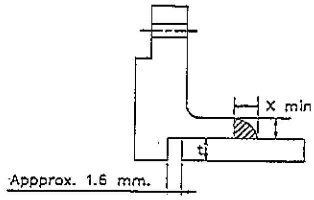
Cx (min.)=1.25t but not less than 3.2 mm.

JOINT No. A1



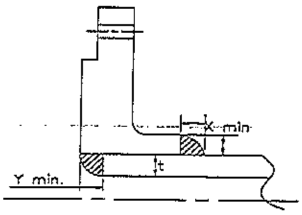
Misalignment (max.) = 1.5 mm.

JOINT No. B1 & B2



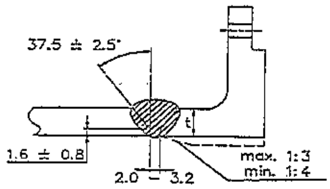
X min. = The lesser of 1.4t or thickness of the hub.

JOINT No. A2



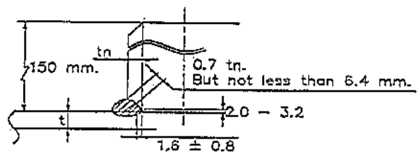
X min. = The lesser of 1.4t or thickness of the hub.
Y min .= The lesser of t or 6.4 mm.

JOINT No. A3



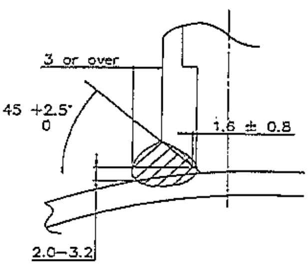
Misalignment (max.) = 1.5 mm.

JOINT No. B3

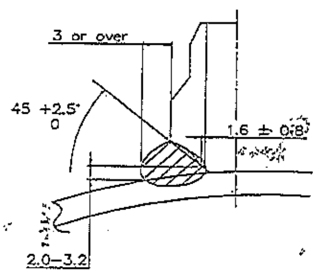


Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

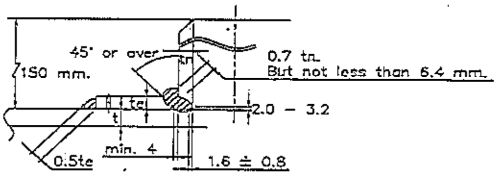
JOINT No. B4



JOINT No. C1

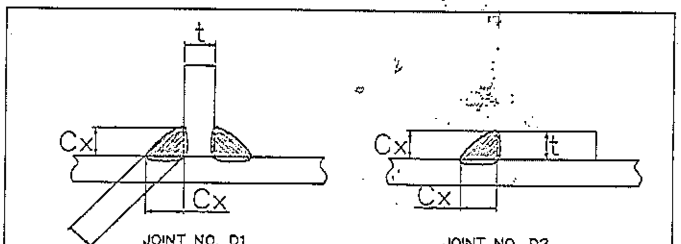


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or 0.5tn.

JOINT No. B5



JOINT NO. D1

JOINT NO. D2

Theoretical throat

Note : Cx = Size of weld as specified
Theoretical throat = 0.707 x Cx

Procedure Qualification Record (PQR)

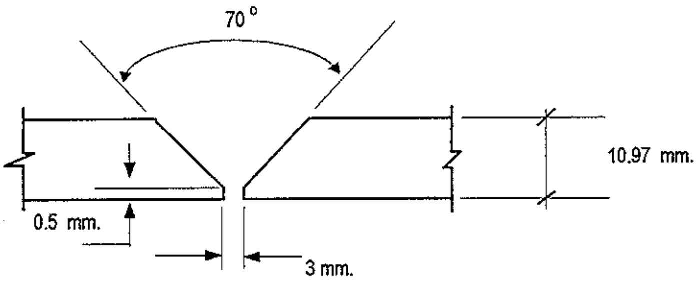
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : KWC-06A0017-PQR-RNC-PN1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	KWC-06A0017-PQR-RNC-PN1	Date :	23 February 2013
WPS Number :	KWC-06A0017-WPS-RNC-PN1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic):	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)			
Material Spec.	A 106	To	A 106
Type or Grade	Gr.B	To	Gr.B
P. No.	1	To	1
Thickness of Test Coupon	10.97 mm.		
Diameter of Test Coupon	ø 6"		
Others			

FILLER METALS (QW-404)	GTAW	SMAW
SFA Specification	A 5.18	A 5.1
AWS Classification	ER70S-G	E7016
Filler Metal F-No.	6	4
Weld Metal Analysis A-No.	1	1
Size of Filler Metal	Ø 2.4 mm.	Ø 3.2 mm.
Others	Kobelco TGS-50	Kobelco LB-52
Weld Metal Thickness	-	

POSITION (QW-405)	
Position of Groove	6G
Progression of Welding (Uphill or Down)	Up Hill
Others	N/A

PREHEAT (QW-406)	
Preheat Temperature	37° C
Interpass Temperature (max.)	188° C
Others	N/A

POSTWELD HEAT TREATMENT (QW-407)	
Temperature	N/A
Hold Time	N/A
Others	

GAS (QW-408)

	Percent Composition		
	Gas (es)	Mixture	Flow Rate
Shielding	Argon	99.99%	8 L/Min.
Trailing	-	-	-
Backing	-	-	-

ELECTRICAL CHARACTERISTICS (QW-409)			
Current	DC (GTAW)	DC (SMAW)	
Polarity	EN	EP	
Amperage	94 - 150 (GTAW)	Voltage	10 - 12 (GTAW)
	100 - 112 (SMAW)		23 - 24 (SMAW)
Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2		
Others	N/A		

TECHNIQUE (QW-410)	
Travel Speed	4.73 - 15.24 cm/min.
String or Weave Bead	Both
Oscillation	N/A
Multiple or single Pass	Multiple Pass
Multiple or Single Electrodes	Multiple Electrodes
Others	N/A

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542
69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number-: KWC-06A0017-PQR-RNC-PN1

Tensile Test Report No.TS-13-02-084

Specimen	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (KN)	Ultimate Strength (N/mm^2)	Type of Failure & Loc.
545/13 TR1	19.05	9.78	186.31	97.06	520.97	OUT OF WELD
545/13 TR2	19.02	9.88	187.92	97.62	519.46	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-033

Type and Figure Number	Type of Bend	Indication	Results
545/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
545/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
545/13 BR1	Transverse root bend	2.12,1.57,1.78 (WELD ZONE)	Accepted
545/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A

Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-009

Hardness Testing Report No. HV-13-02-015

Macrostructure Report No. MA-13-02-049

Impact Testing Report No.IM-13-02-092

Welders Name Mr.Prathom M. Welder Number W-009

Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor : QA/QC RNC

Date : 23/02/13

Inspected by : Mr.Wichit K.

Certified by : Mr.Chaloemkiet R.

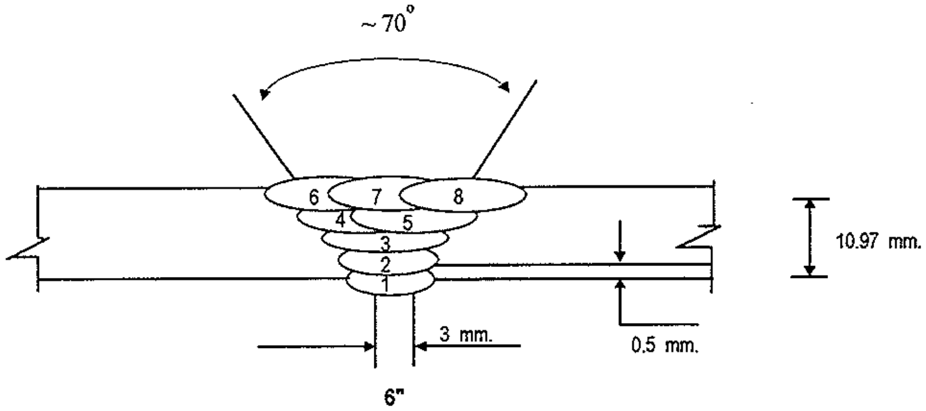
Date : 22 February 2013

PAE Technical Service Company Limited Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : KWC-06A0017-PQR-RNC-PN1
		WPS No. : KWC-06A0017-WPS-RNC-PN1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for KWC-06A0017-PQR-RNC-PN1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or Weave Bead	Wire Feed Speed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	94	10	DCEN	-	-	4.73	11.93	8	-
2	GTAW	ER70S-G	2.4	150	12	DCEN	-	-	8.88	12.17	8	-
3	SMAW	E7016	3.2	100	23	DCEP	-	-	11.71	11.79	-	-
4	SMAW	E7016	3.2	110	24	DCEP	-	-	15.24	10.40	-	-
5	SMAW	E7016	3.2	111	24	DCEP	-	-	9.51	16.81	-	-
6	SMAW	E7016	3.2	110	24	DCEP	-	-	11.68	13.57	-	-
7	SMAW	E7016	3.2	112	24	DCEP	-	-	9.51	16.96	-	-
8	SMAW	E7016	3.2	111	24	DCEP	-	-	7.39	21.63	-	-

PAE Technical Service Company Limited

Public Registration No. (5) 1571/2542

69 Srinakarin Rd., suanluang, Bangkok 10250, Thailand. Tel (662) 322-022 Fax : (622) 721-2577

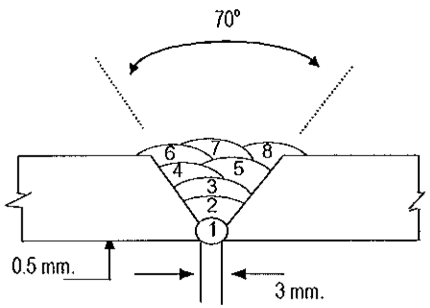
PAE TECHNICAL SERVICE CO.,LTD
NDT, TESTING, INSPECTION & ENGINERING CONSULTANT
69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250
Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

WELDER PROCEDURE QUALIFICATION RECORD

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-017
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : KWC-0610017-WPS-RNC-PN1	Date : 13-Feb-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode	Results	
					Spec	Size	Specification	Visual	RT
W-009	MR.PRATHOM M.	GTAW / SMAW	6G	ASME IX	A106 Gr.B	ø 6" Thk. 10.97 mm.	A5.18 / A5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	94	10	4.73	37	11.93	Argon	8	-	-
2	GTAW	ER70S-G	2.4	DCEN	150	12	8.88	119	12.17	Argon	8	-	-
3	SMAW	E7016	3.2	DCEP	100	23	11.71	123	11.79	-	-	-	-
4	SMAW	E7016	3.2	DCEP	110	24	15.24	143	10.40	-	-	-	-
5	SMAW	E7016	3.2	DCEP	111	24	9.51	150	16.81	-	-	-	-
6	SMAW	E7016	3.2	DCEP	110	24	11.68	153	13.57	-	-	-	-
7	SMAW	E7016	3.2	DCEP	112	24	9.51	180	16.96	-	-	-	-
8	SMAW	E7016	3.2	DCEP	111	24	7.39	188	21.63	-	-	-	-

PAE Inspection By	Witness By	Approved By
SIGNATURE :	SIGNATURE :	SIGNATURE :
NAME : Mr.Wichit K.	NAME : Mr. Chatchawan S.	NAME : ภูชิน สีหอมไชย
DATE : 14-Feb-2013	DATE : 19/02/13	DATE : Phuchin Seehomchai

PAE TECHNICAL SERVICE

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Testing Laboratory LPN Plate Mill Public Company Limited

LPN PLATE MILL

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

29780

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-084

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 6" (10.97 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm^2)	Gauge Length (mm)	Load (kN)	Strength (N/mm^2)	Load (kN)	Strength (N/mm^2)		
1	545/13 TR1	N/A	9.78	19.05	186.31	N/A	N/A	N/A	97.06	520.97	N/A	OUT OF WELD
2	545/13 TR2	N/A	9.88	19.02	187.92	N/A	N/A	N/A	97.62	519.46	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. RNC QA/QC RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

20786

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1

Report No. BD-13-02-033

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" (10.97 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	545/13 BF1	Transverse face bend	10.01x38.10x162.51 mm	No Open Discontinuity	N/A	-
2	545/13 BF2	Transverse face bend	10.00x38.05x161.70 mm	No Open Discontinuity	N/A	-
3	545/13 BR1	Transverse root bend	10.25x38.05x163.03 mm	Open Discontinuity	WELD ZONE	2.12, 1.57, 1.78 mm
4	545/13 BR2	Transverse root bend	10.02x37.97x161.18 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

TESTING No. 0134

9792

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-092

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a$^{\varepsilon1}$
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" (10.97 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	19 February 2013	Equipment 's Capacity	542 Joules
Tested Date	20 February 2013	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Specimen Dimension Thickness (mm)	Width (mm)	Length (mm)	Results Energy Absorbed (Joules)	Average (Joules)	Location of Test	Remark
1	545/13-1	-46°C	7.507	10.007	54.81	147.11	199.65	Weld Metal	-
2	545/13-2	-46°C	7.505	10.007	54.79	262.94			
3	545/13-3	-46°C	7.506	10.008	54.86	188.90			
4	545/13-1	-46°C	7.506	10.009	54.92	7.52	9.51	Base Metal	-
5	545/13-2	-46°C	7.506	10.008	54.95	11.32			
6	545/13-3	-46°C	7.506	10.006	54.96	9.70			
7	545/13-1	-46°C	7.507	10.006	54.85	91.68	90.00	HAZ	-
8	545/13-2	-46°C	7.507	10.008	54.88	93.30			
9	545/13-3	-46°C	7.505	10.008	54.78	85.02			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009

RNC RENEW COILS ENGINEERING & SUPPLY CO., LTD.
QA/QC
RNC

Reported by : ....

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : ....

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th 13696

Macrostructure Analysis Report

Report No. MA-13-02-049

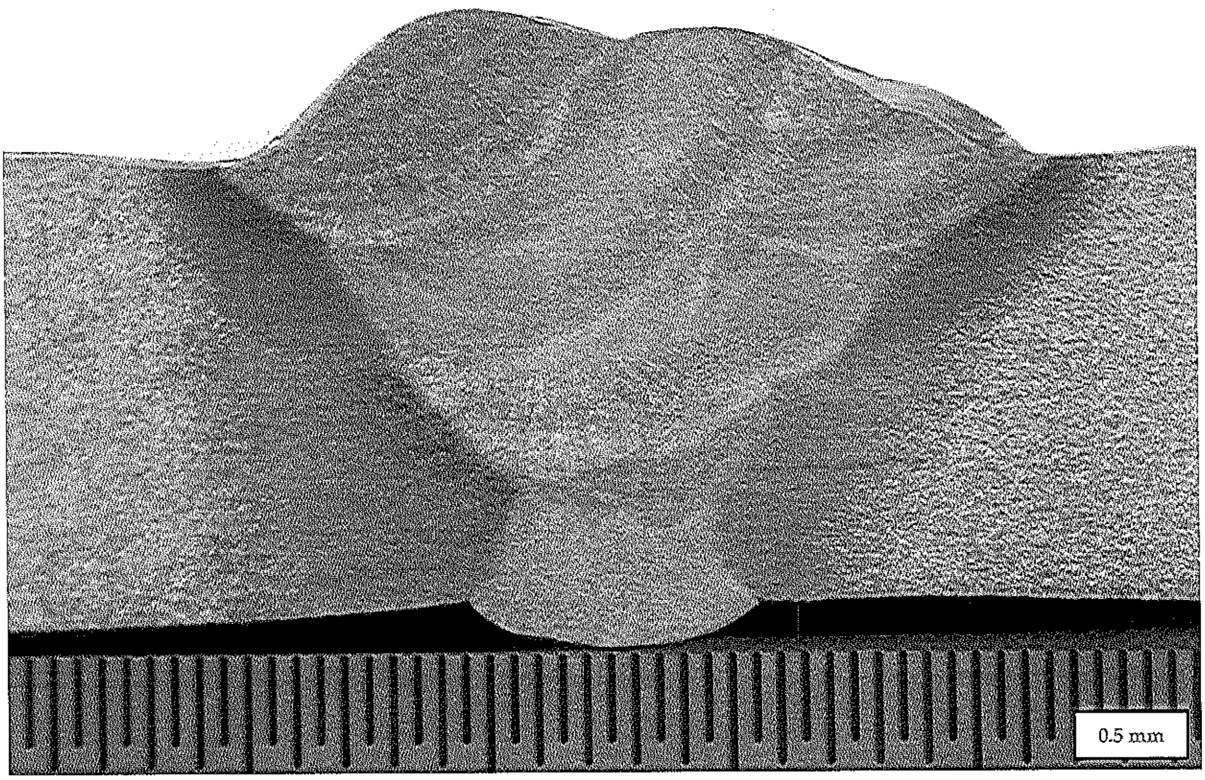
Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1
Test No.	545/13 MA
Test Location	-
Material Specification	A106 Gr.B
Dimension	6" (10.97 mm Thk.)
Welding Process/Position	GTAW+SMAW/6G
Type of Joint	Butt Jiont
Welder Name / No.	Mr. Prathom M. / W-009
Date Received	18 February 2013

Material Characterization	Pipe
Equipment used	Microscope upto 40X (times)
Serial No.	LEICA 0650854912
Method of Preparation	ASME Secction IX : 2010 edition
Test Method	ASME Secction IX : 2010 edition
Standard of Test	ASME Secction IX : 2010 edition
Etchant Reagent	25 ml HNO_3 + 75 ml H_2O
Time of Etching	5 minutes
Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

Additional details : -

Reported by : N.

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 22/02/13

RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 22/02/13

LPN PLATE MILL

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

13232

Hardness Testing Report

Report No. HV-13-02-015
Revision No. 0

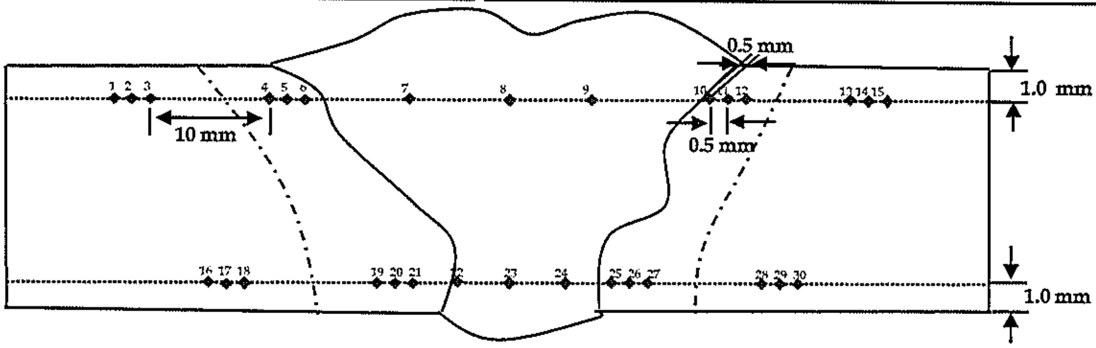
Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1
Test No.	545/13 HN
Project Name	WELDER SUPPLY AND PQR
Project No.	N/A
Material Specification	A106 Gr.B
Dimension	6" (10.97 mm Thk.)
Welding Process/Position	GTAW+SMAW/6G
Type of Joint	Butt Jiont
Date Received	18 February 2013
Date Tested	20 February 2013

Material Characterization	Pipe
Test Method	ASTM E92-82 (Reapproved 2003)$^{\epsilon 2}$
Standard of	ASME Secction IX : 2010 edition
Type of test	Vicker Hardness Test
Equipment	Wilson Wolpert , 432 SVD (V32D219)
Indentor Size	Pyramid diamond, angle of 136°
Test Load	HV 10 Kgf.
Total Force Dwell Time	15 s
Cal. Block	No. A60023 No. A62072
Ambient Temperature	26.0 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	142.0	16	BASE	148.5		**************************	
2	BASE	141.3	17	BASE	147.5			
3	BASE	140.8	18	BASE	148.3			
4	HAZ	147.8	19	HAZ	180.4			
5	HAZ	160.8	20	HAZ	176.7			
6	HAZ	175.0	21	HAZ	176.2			
7	WELD	180.4	22	WELD	170.0			
8	WELD	166.0	23	WELD	164.7			
9	WELD	164.7	24	WELD	167.0			
10	HAZ	180.8	25	HAZ	175.9			
11	HAZ	164.5	26	HAZ	176.9			
12	HAZ	155.9	27	HAZ	178.6			
13	BASE	139.5	28	BASE	143.9			
14	BASE	139.9	29	BASE	141.9			
15	BASE	138.0	30	BASE	142.7			

Additional details : -

Reported by : ......
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 22/02/13

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD.
QA/QC
RNC

Approved by : ......
(Mr.Pavaret Preedawiphat)
Deputy Managing Director
Date : 22/02/13

LPN PLATE MILL

Notes
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ArcelorMittal South Africa Limited
Tubular Products
Genl. Hertzog rd.
P O Box 48 Vereeniging 1930
South Africa

MATERIAL TEST CERTIFICATE
SEAMLESS TUBE

Telephone +27 (0)16 450 4220
Fax +27 (0)16 423 4906

Licensed under API Spec 5L and 5CT
5L - 0319
5CT - 0419

ISO 9001:2000
DE-395997 QM
Test Certificate EN 10204:2004 TYPE 3.1

ArcelorMittal

Customer:
Order No: 4000010387
Certificate Reference No: 040001591051
Product: FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES
Specification: ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/A53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04
Product Marking: ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 07-10 ASTM/ASME A/SA106B A/SA53B 168.300 (6.626) 10.970 (0.432) 6.000 (19.700) L245/B L290/X42 PSL1 SMLS TESTED 20.7 MPa (3000 psi) CAST NO:89B7130 PROD/O NO: T6641038710

Customer Order/Contract No: SSG7 132443
Material No: 1000000468
Cast/Heat No: 89B7130

General Information

6" XS|S80

Quantity	Mass	Dimensions			Total Length
		Tube OD	Thickness	Length	
90	22,983.300(kg)	168.300(mm)	10.970(mm)	6.000(m)	540.000(m)
	50,668.983(lb)	6.626(")	0.432(")	19.700 (ft)	1,773.000(ft)

Steel making process	Final Rolling Operation
Electric Arc	Final hot rolling operation finished above 860°C and cooled in still air.

Chemical Composition

Element(%)	C	Si	Mn	S	P	Cr	Ni	Mo	Cu	V	Al	Ti	Sn	Ca	N	B	Nb	CE
Minimum	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Maximum	0.200	-	1.30	0.030	0.030	0.50	0.50	0.150	0.500	-	-	-	-	-	-	-	-	0.41
Heat	0.170	0.31	0.82	0.008	0.019	0.16	0.08	0.016	0.110	0.005	0.032	0.001	0.008	0.0012	0.0027	0.0003	0.0020	0.36
Product	0.170	0.310	0.820	0.008	0.019	0.160	0.080	0.016	0.110	0.005	0.032	-	-	-	-	-	-	0.356
Product (ADD)	0.170	0.310	0.820	0.008	0.019	0.160	0.080	0.016	0.110	0.005	0.032	-	-	-	-	-	-	0.356

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

Mechanical Properties

Specification Limits	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
Minimum	415	60000	290	42000	30.0
Maximum	-	-	-	-	-
(1) Actual	503	72954	347	50328	38.0
(2) Actual	507	73534	349	50618	38.0
(3) Actual					
(4) Actual					

	UTS (Rm) MPa	UTS (Rm) psi	Yield (0.5%) MPa	Yield (0.5%) psi	% EL 50mm
(5) Actual					
(6) Actual					
(7) Actual					
Orientation & type of tensile test	Longitudinal, Strip				
Width of tensile piece (mm)	38.4mm				
Orientation of impact test piece					

OTHER TESTS	
Flattening	Passed
Hydrostatic	20700 kPa (3000 psi) - 5 sec
NDI: EMI	PASS - ASTM E570 - 12.5% NOTCH
NDI: UT	UT not required
HV 10 Kg	153 154 156

CERTIFY TRUE COPY
BY URMENA LIMITED
PURCHASE ORDER NO. 80-0045017873
ITEM NO. 6

ptt GLOBAL CHEMICAL
Chayut S.
Inspection QA/QC Division
Integrity and Reliability
14/2/2013

Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2007. Dimensions to ANSI B36.10M-2004
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2007 add 2009
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: R. Bester
Date of Issue: 2010.08.07
Certified by: 6

We hereby certify that the steel grade and quality level of all products are in conformity with the order and comply fully with specification requirements. No changes, amendments or additions may be made to this document. Any changes which are effected shall invalidate this certificate.

UR MENA LIMITED Q.A. TEL 0-2138-8200

20/-17 LS4

KOBELCO

INSPECTION CERTIFICATE

PURCHASER

Date of Issue: 11/Aug/2011
Certificate No.: TF11-035
Date of Inspection: 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
LB-52	3.2 x 350	12110	AWS A5.1 E7016

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97

Tensile Test of Weld Metal			Impact Test of Weld Metal			
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009
465 MPa	554 MPa	33 %	-30 °C	Avg.	246 J	

Welding Conditions		Postweld Heat Treatment	REMARKS
Type of Current	AC	--- °C x --- Hr.	*1* : Mn+Ni+Cr+Mo+V
Welding Current	120 A		
Welding Voltage	24 V		

RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD. QA/QC RNC

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

KOBELCO

INSPECTION CERTIFICATE

Date of Issue: 17/Feb/2011
Certificate No.: KF11-006
Date of Inspection: 15/Feb/2011

PURCHASER

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
TG-S50	2.4	F0B2704	AWS A5.18 ER70S-G

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V
WIRE	0.10	0.73	1.39	0.011	0.017	0.13	0.01	0.01	< 0.01	0.002

Tensile Test of Weld Metal			Impact Test of Weld Metal		
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy	
453 MPa	548 MPa	32 %	-30 °C	Avg.	137 J

Tested size for Mechanical test: 2.4 mm
Tested MFG. No. of Mechanical test: F9B4140

Welding Conditions

Type of Current	DCEN	Shielding Gas	Ar
Welding Current	238 A		
Welding Voltage	14 V		

REMARKS

QA/QC RNC
RNC RENEW COILS ENGINEERING & SUPPLY CO.,LTD.

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

KOBE MIG WIRE (THAILAND) CO., LTD.

CHIEF INSPECTOR

PAGE1

KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Welding Procedure Specification (WPS)

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name RENEW COILS ENGINEERING LIMITED PARTNERSHIP
WPS No. KWC-06A0017-WPS-RNC-PN1 Ref. PQR No. KWC-06A0017-PQR-RNC-PN1
Revision No. 0 Date. JULY 20, 2006
Welding Process(es) GTAW and GMAW Type(s) MANUAL
(Automatic, Manual, Machine, or Semi-auto)

SMAW T.C.

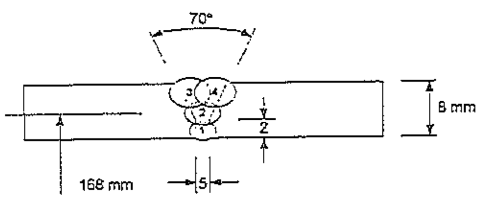
JOINTS (QW - 402)
Joint Design 6G
Backing (Yes) ____ (No) ✓
Backing Materials (Type) -
(Refer to both backing and retainers)

☐ Metal ☐ Nonfusing metal
☐ Non-metallic ☐ Other

Sketches, Production Drawings, Weld Symbols or Written Description should show the general arrangement of the parts to be welded. Where applicable, the root spacing and the details of weld groove may be specified.

(At the option of the Mfgr., sketches may be attached to illustrate join design, weld layers and bead sequence, e. g., for notch toughness procedures, for multiple process procedures, etc.)

DETAILS



***BASE METALS (QW - 403)**
P No. 1 Group No. 1 To P No. 1 Group No. 1
OR
Specification type and grade A106 Gr B
to Specification type and grade A106 Gr B
OR
Chem. Analysis and Mech. Prop. -
to Chem. Analysis and Mech. Prop. -
Thickness Range:
Base Metal: Groove 3-16 mm Fillet -
Other - -

***FILLER METALS (QW - 404)**

Spec. No. (SFA)	5.18	5.1
AWS No. (Class)	ER 70S-6	E 7016
F-No	6	4
A-No	1	1
Size of Filler Metals	Ø2.4mm	Ø3.2mm
Weld Metal		
Thickness Range:	4 mm (max)	12 mm (max)
Groove	4 mm (max)	12 mm (max)
Fillet	All	All
Electrode-Flux (Class)	-	-
Flux Trade Name	-	-
Consumable Insert	-	-
Other	-	-

* Each base metal-filler metal combination should be recorded individually.

(R. Phaonium) Designed
(S. Peansukmanee) Approved
Authorized Manufacturer

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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER

Welding Procedure Specification (WPS) Cont'd

WPS (Back)

WPS No. KWC-06A0017-WPS-RNC-PN1 Rev. 0

POSITIONS (QW - 405)
Position(s) of Groove All
Welding Progression: Up ✓ Down
Position(s) of Fillet -

POST WELD HEAT TREATMENT (QW - 407)
Temperature Range -
Time Range -

PREHEAT (QW - 406)
Preheat Temp. Min. -
Interpass Temp. Max. 100°C
Preheat Maintenance -
(Continuous or special heating where applicable should be recorded)

GAS (QW - 408)

	Percent Composition Gas(es)	Mixture(s)	Flow Rate
Shielding	Ar	-	20 lpm
Trailing	-	-	-
Backing	Ar	-	20 lpm

ELECTRICAL CHARACTERISTICS (QW - 409)
Current AC or DC DC Polarity GTAW DC-, SMAW DC-
Amps (Range) GTAW 80, SMAW 100 Volts (Range) GTAW 13.2, SMAW 32.2
(Amps and volts range should be recorded for each electrode size, position, and thickness, etc. This information may be listed in a tabular form similar to that shown below.)
Tungsten Electrode Size and Type 2% Thotiated (EWTh-2) (Pure Tungsten, 2% Thoriated, etc.)
Mode of Metal Transfer for GMAW - (Spray arc, short circuiting arc, etc.)
Electrode Wire feed speed range -

TECHNIQUE (QW - 410)
String or Weave Bead GTAW STRING, SMAW WEAVE
Orifice or Gas Cup Size Ø10 mm
Initial and Interpass Cleaning (Brushing, Grinding, etc.) Brushing and Grinding
Method of Back Gouging -
Oscillation -
Contact Tube to Work Distance -
Multiple or Single Pass (per side) MULTIPLE (4)
Multiple or Single Electrodes SINGLE
Travel Speed (Range) GTAW 4-10 cm/min, SMAW 5-15 cm/min
Peening -
Other

Weld Layer(s)	Process	Filler Metal Class	Filler Metal Dia. (mm)	Current Type Polar.	Current Amp. Range (A)	Volt Range (V)	Travel Speed Range (cm/min)	Others (e.g., Remarks, Comments, Hot Wire Addition, Technique, Torch Angle, Etc.)
1	GTAW	ER70S6	2.4	DCEN	80-110	13.2-15.5	4-10	
2	SMAW	E 7016	3.2	DCEN	100-150	24-31	5-15	
3	SMAW	E 7016	3.2	DCEN	100-150	24-31	5-15	

(R. Phaonium)
Designed

(S. Peansukmanee)
Approved

Authorized
Manufacturer

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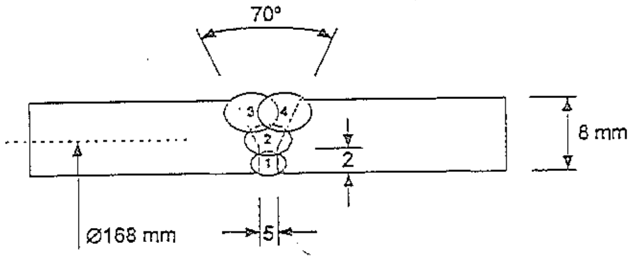
KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Procedure Qualification Records (PQR)

PROCEDURE QUALIFICATION RECORDS (PQR)

Company Name RENEW COILS ENGINEERING LIMITED PARTNERSHIP
Procedure Qualification No. KWC-06A0017-PQR-RNC-PN1 Date JULY 20, 2006
Pre-Qualified WPS No. KWC-06A0017-WPS-RNC-PN1 Revision No. 0
Welding Process(es) GTAW+SMAW
Types (Manual, Automatic, Semi-Auto.) MANUAL

JOINTS (QW – 402)



Groove Design of Test Coupon
(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used.)

BASE METALS (QW – 403)
Material Spec. A106
Type or Grade Gr. B
P No. 1 to P No. 1
Thickness of Test Coupon 8 mm
Diameter of Test Coupon 168 mm
Other(s) PIPE

FILLER METALS (QW – 404)
SFA Specification 5.18 5.1
AWS Classification ER70S-6 E7016
Filler Metal F-No. 6 4
Weld Metal Analysis A-No. 1 1
Size of Filler Metal Ø2.4mm Ø3.2mm
Other -
Weld Metal Thickness 8 mm

POSITION (QW – 405)
Position of Groove 6G
Weld Progression (Up, Downhill) UPHILL
Other -

PREHEAT (QW – 406)
Preheat Temp. -
Interpass Temp. NOT EXCEED 100°C
Other -

POSTWELD HEAT TREATMENT(QW–407)
Temperature -
Time -
Other(s) -

GAS(QW–408) Percent Composition

	Gas(es)	Mixture	Flow Rate
Shielding	Ar	-	20 lpm
Trailing	-	-	-
Backing	Ar	-	20 lpm

ELECTRICAL CHARACTERISTICS(QW–409)
Current DIRECT CURRENT
Polarity GTAW DC- SMAW DC-
Amp 80 100 Volt 13.2 32.2
Tungsten Electrode Size 2.4 mm
Other -

TECHNIQUE(QW–410) GTAW SMAW
Travel Speed 4-10 cm/min 5-15 cm/min
String or Weave Bead STRING WEAVE
Oscillation -
Multi or Single Pass (per side) MULTI PASS
Single or Multiple Electrode SINGLE
Other -

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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Procedure Qualification Records (PQR) Cont'd

PROCEDURE QUALIFICATION RECORDS (BACK)

TEST RESULTS OF TEST COUPON PQR No. KWC-06A0017-PQR-RNC-PN1

TENSILE TESTS (QW – 150)

Specimen No.	Width (mm)	Thickness (mm)	Area (mm^2)	Ultimate Total Load (kN)	Ultimate Unit Stress (MPa)	Type of Failure & Location
PN1-TS01	22.00	4.90	107.80	69.550	645.189	Weld Area
PN1-TS02	22.90	5.60	128.24	72.350	564.176	Weld Area

GUIDED BEND TESTS (QW – 160)

Type & Figure No.	Result	Remarks
PN1-FB1, Face Bend	No Open Defect at Weldment	-
PN1-FB2, Face Bend	Open Discontinuity <3.2 mm at Weld	-
PN1-RB1, Root Bend	No Open Defect at Weldment	-
PN1-RB2, Root Bend	Open Discontinuity <3.2 mm at Weld	-

TOUGHNESS TESTS (QW – 170)

Specimen No.	Notch Location	Specimen Size	Test Temp.	Impact Values Ft. Lbs.	Shear	Mils	Drop Weight Break (Y/N)

Comments: __________

FILLET WELD TESTS (QW – 180)

Result-Satisfactory: YES____ NO____ Penetration into Parent Metal: YES____ NO____

Macro Result: __________

OTHER TEST

Type of Test __________

Deposit Analysis __________

Other __________

Welder's Name __________ Clock No.______ Stamp No.______

Test Conducted by KINGWELD Laboratory Test No. KWC-06A0017

We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.

Ekkachai Warinsiriruk
(E. Warinsiriruk)
Inspector

Authorized
Manufacturer

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